

Divya Rana

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Education

Rajiv Gandhi Institute of Petroleum Technology (RGPT)

Bangalore, India

Integrated BTech(CSE) + MTech(AI)

2022–2027

Institution of National Importance (IIT-Equivalent Status)

M.Tech Research Focus: *Adaptive Fusion for Cross-Dataset Generalization in Deepfake Detection: A Reenactment-Based OOD Evaluation*

Professional Experience

Roostoo Labs

Platform Enhancement and Technical Research Intern

May 2025 – Aug 2025

- Engineered automated decision-making and signal processing pipelines using Python, analyzing large-scale time-series interaction data to model user behavior patterns and latent dynamic trends.
- Designed and deployed API integration layers and custom frontend analytics tools to evaluate real-time behavioral metrics, model latency, and system performance indicators under concurrent load.
- Conducted root-cause defect analysis and statistical evaluations in an Agile engineering setup, optimizing pipeline execution efficiency and ensuring high data reliability across endpoints.

Old Skool Labs

Full Stack Developer Intern

May 2024 – Aug 2024

- Architected and integrated RESTful APIs and responsive user interfaces, optimizing user interaction flows and dynamic data retrieval speeds across client-facing web platforms.
- Formulated data validation routines and SQL-based backend procedures, reducing query response latencies and enabling seamless cross-platform data synchronization.
- Collaborated in cross-functional Agile workflows using Git-based version control, participating in sprint planning, automated testing, and release verification cycles.

Technical Projects

Deep Thinking Multi-Agent Behavioral Trading & Causal Evaluator

Python, PyTorch, LangGraph, LLMs, SQL

- Architected a distributed multi-agent state-machine system utilizing PyTorch and LLMs to model sequential decision trajectories and complex user interaction behaviors across dynamic market conditions.
- Formulated a causal evaluation framework combining counterfactual reasoning and LLM-as-Judge metrics to quantify user policy interventions and retention impacts under real-time telemetry streams.

Sequential User Behavior & Content Recommendation Engine

Python, PyTorch, SQL, Transformers, AWS

- Developed a deep representation learning pipeline analyzing large-scale user interaction logs and sequential viewing histories to predict engagement and user satisfaction metrics.
- Executed SQL ETL routines over multi-gigabyte interaction datasets and leveraged PyTorch transformer architectures to evaluate counterfactual intervention strategies for personalized content delivery.

Real-Time Affect & Facial Emotion Recognition System

Python, PyTorch, OpenCV, DeepFace, Edge AI

- Built a real-time visual representation model leveraging deep convolutional networks in PyTorch for continuous user affect recognition and engagement state classification.
- Optimized model inference pipelines for edge deployment, achieving high frame-rate feature extraction and accurate classification of fine-grained emotional responses.

Technical Skills

Languages: Python (Advanced), C++, Java, SQL (Complex Queries, Joins, Aggregations), JavaScript, C, MATLAB

AI / ML & Causal Inference: PyTorch, Deep Learning, LLMs (Fine-Tuning, Transformers), Causal Inference, Counterfactual Reasoning, Representation Learning, Sequential Modeling, Recommendation Systems

Data & Infrastructure: SQL (MySQL, PostgreSQL), Pandas, NumPy, AWS (S3, Athena, EC2), Vector Databases (Pinecone, FAISS), REST APIs, Docker, Linux, Git/GitHub

Frameworks & Tools: Flask, FastAPI, LangChain, OpenCV, DeepFace, Jupyter Notebooks, VS Code

Core Concepts: Data Structures & Algorithms (DSA), User Behavior Modeling, Machine Learning Theory, Model Evaluation, System Design

Achievements & Leadership

- KodeKurrent 2025 Hackathon:** Secured 2nd Runner-Up position at the annual flagship hackathon organized by IEEE-RGPT for designing and implementing an automated decision and algorithmic modeling system.
- Dark Pattern Buster Hackathon (DPBH) 2023:** Selected as Runner-Up for building ML-driven behavioral analysis tools identifying deceptive user interface patterns in web platforms.
- Technical Research & Content Development:** Authored structured technical analyses on algorithmic optimization and machine learning strategies for technical submission and peer review.
- Academic Mentorship & Leadership:** Active peer mentor at RGPT, guiding undergraduate students in Data Structures, Machine Learning fundamentals, and Python software development.